**Abbreviations**

| | |
|---|---|
| cao | correct answer only |
| dep | dependent |
| FT | follow through after error |
| isw | ignore subsequent working |
| oe | or equivalent |
| SC | Special Case |
| nfww | not from wrong working |
| soi | seen or implied |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 1 | 607 532 | **1** | |
| 2(a) | 64 | **1** | |
| 2(b) | 65 | **1** | |
| 2(c) | 62 | **1** | |
| 2(d) | 61 67 cao | **2** | **B1** for each<br>or for two correct and one extra |
| 3 | A congruent triangle drawn | **1** | |
| 4(a) | 6 | **1** | |
| 4(b) | 19 | **1** | |
| 5 | 270 ÷ 25 | **M1** | |
| | [=] 540 ÷ 50<br>[=] 1080 ÷ 100<br>[=] 10.8[0] | **A1** | |
| 6 | Bars are not the same width oe<br>Frequency scale is not linear oe | **2** | **B1** for each. |
| 7(a) | $6 \times (7 - 5) + 4 = 16$ | **1** | |
| 7(b) | $(-2)^2 + 24 \div 12 - 4 = 2$ | **1** | |
| 8 | 13 or −13 | **1** | |
| 9(a) | 54 | **2** | **M1** for $\frac{360}{8+5+4+3}$[×3] or $\frac{3}{8+5+4+3}$[×360] oe |
| 9(b) | Yes [because] the proportion of red cars will be smaller oe | **1** | |
| 10(a) | $\begin{pmatrix} 3 \\ 4 \end{pmatrix}$ | **1** | |

| Question | Answer | Marks | Partial Marks |
|---|---|---|---|
| 10(b) | $\begin{pmatrix}12\\48\end{pmatrix}$ | **1** | |
| 11 | 180 | **3** | **M2** for $[2\times](8\times6+8\times3+3\times6)$ oe<br><br>or **M1** for 8 × 6 or 8 × 3 or 3 × 6 |
| 12(a) | $\frac{d}{n}$ oe | **1** | |
| 12(b) | $20 - rx - ay$ oe | **2** | **B1** for $rx$ and $ay$ seen and not spoilt |
| 13(a) | 102 | **1** | |
| 13(b) | 18.94 cao | **2** | **B1** for 18.9 or 18.93 or 18.935…<br>If zero scored, **SC1** for rounding *their* value correctly to 2 decimal places |
| 14 | 1 : 4 : 60 | **2** | **B1** for 0.5[:2:]30 oe |
| 15(a) | 13, 16, 21 | **2** | **B1** for 2 correct terms in correct position<br>or **SC1** for 12, 13, 16 |
| 15(b) | Yes<br>$n = 72$ | **2** | **M1** for $n^2 + 12 = 5196$ or better<br>or for a correct worded description for finding $n$<br>or for 5184 is a square number |
| 16 | $\pi \quad \sqrt{3}$ cao | **1** | |
| 17 | 10 cao | **1** | |
| 18 | $\frac{30-2^2}{800\times0.05}$ | **M1** | |
| | 0.65 | **A1** | If **M0** scored, **SC1** for 3 correct values or all correct but with any trailing zeros |
| 19 | 30.65, 30.75 | **2** | **B1** for each<br>or if zero scored, **SC1** for both answers correct but reversed |
| 20(a) | $6a - 4b$ final answer | **2** | **B1** for $6a$ or $-4b$ in final answer<br>or for $6a - 4b$ spoilt |
| 20(b) | $x(x - 8y)$ final answer | **1** | |